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Coalescence-induced jumping of droplets on superhydrophobic substrates with a beam structure

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ABSTRACT

Coalescence-induced jumping motion of droplets on superhydrophobic surfaces has potential applications for anti-icing, self-cleaning and anti-fogging, etc. In the present study, the coalescence-induced jumping of mesoscale water droplets on substrate without and with beam is investigated by using many-body dissipative particle dynamics (MDPD) simulations. For the cases of jumping on flat substrate without beam, the jumping velocity scaled by the inertial-capillary velocity (v_{ic}) is nearly constant at 0.105 for droplet radius varying from 116.04 nm to 290.10 nm. For coalescence-induced jumping on substrate with a beam, the jumping velocity can be enhanced to $0.7v_{ic}$, and the jumping is accompanied by the merging process, which is different from that of the droplets merging on substrate without beam. The stages of merging on substrate with a beam can also be divided into 4 stages: I. expansion of liquid bridge; II. detaching from substrate; III. jumping away from beam; IV. moving freely in air. The conversion efficiency of jumping kinetic energy increases with the beam height in the range from zero to the radius of droplet. Hydrophobic and hydrophilic beams lead to coalesced droplets standing on the beam or wrapping the beam. Our work can provide new insights in the droplets coalescence dynamics, and help design water resource collection system and microfluidics devices.

1. Introduction

When two droplets merge on a low-adhesion surface, the coalesced droplet jumps away from the surface without any external energy. This self-propelled phenomenon is defined as coalescence-induced droplet jumping, which was firstly reported by Kollera and Grigull about the experiments of condensing mercury droplets bouncing in 1969 [1]. In recent years, studies have found that the spontaneous jumping behavior of droplets has a great potential applied in enhancing performance of anti-icing and anti-frosting surface [2–6], self-cleaning [7–10], condensing heat transfer and energy collection [11–16] etc. Thus, the mechanism of coalescence-induced droplet jumping has been attracted lots of attention.

The behavior of coalescing droplet depends on the balance of released surface energy, work of adhesion, kinetic energy and viscous dissipative energy. Besides, for droplets whose radius is larger than the capillary length, the gravitational potential energy is also taken into account. The capillary-inertial velocity is defined as $v_{ic} = \sqrt{\gamma/\rho r}$, where γ , ρ and r are the surface tension, density and initial radius of droplet,

respectively. For two inviscid droplets of the same size merging on a superhydrophobic surface with 180° contact angle, if the conversion rate of released surface energy to jumping kinetic energy is 100%, the jumping velocity reaches $1.1v_{ic}$. However, in fact, just a small fraction of released surface energy is converted into the jumping kinetic energy. Previous experiment observations [8,17–19] have demonstrated that the ratio of jumping velocity to capillary-inertial velocity is around 0.2 for droplet radius over a dozen micron, which has been also proved by different numerical simulations including volume of fluid (VoF) [20], lattice Boltzmann method (LBM) [21–24], moment of fluid (MoF) [25] and phase-field method [26]. Wang et al. [27] theoretically explicated that the behavior of droplet jumping can only occur within a certain size ranging from several microns to a few millimeters. However, the critical radius below which the merged droplet is not able to jump away is difficult to confirm. Boreyko and Chen [17] experimentally reported that the critical diameter is of order 10 μm . Cha et al. [28] reduced the critical radius to 500 nm by observing the jumping motion of water droplets on CNT surfaces. Furthermore, the molecular dynamics

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(MD) simulations are employed to investigate the coalescence-induced jumping of nanoscale droplets [29,30]. The simulation results proved that not only nanoscale Ar droplets could jump away from the Au surface but also the jumping velocity still follows the capillary-inertial scaling law as $v_j \sim 0.126v_{ic}$ [31,32]. Above-mentioned works indicated that the energy conversion efficiency is less than 6%, where the low efficiency was attributed to strong viscous dissipation and adhesion of substrate, and it becomes a major problem of coalescence-induced jumping.

The merging of three or more droplets has been demonstrated that the jumping velocity can be improved [10,23]. Yanagisawa et al. [33] reported that the jumping height and the jumping frequency can be increased under an electric field. Researchers have proved that coalescence on a curved surface [34] and on the corner of V-groove [35] enhance the jumping motion. Furthermore, Wang et al. [36] compared the coalescence of two droplets on a rectangular groove with a triangular prism, drawing a conclusion that the triangular prism promotes and the groove restrains the jumping motion. Vahabi et al. [37] studied experimentally and numerically the coalescence of water droplets on a superomniphobic surface with a ridge. The energy efficiency enhances about 570% compared with flat superhydrophobic surface. Xie et al. [38] reported that the jumping velocity enhanced because of the great capillary pressure of the gas-liquid interface in the groove, in which the groove is narrow enough to provide a great counterforce. Peng et al. [39] experimentally demonstrated that not only the macrogroove structure but also the coalescence between deformed and undeformed droplets could enhance the jumping velocity and energy conversion efficiency of coalescence-induced jumping of droplets.

The process and details of coalescence-induced jumping of droplets in mesoscopic scale is difficult to observe in experiments. Mesh-based methods and MD method are not suitable for studying the complex flow in the mesoscopic scale. As a particle-based Lagrangian bottom-up mesoscopic approach, both time and spatial scale for many-body dissipative particle dynamics (MDPD) method are larger than MD method. Besides, the liquid-vapor interface of MDPD is consisted by particles without using any additional interface tracking technology, which is different from mesh-based method. Moreover, by modifying coefficients of conservative force, the collective behavior of liquid and the interaction between liquid-solid can be easily changed. MDPD method has been widely used to investigate multiphase flow dynamics in various complex systems [40–43]. Thus, in this paper, we employed many-body dissipative particle dynamics (MDPD) simulations to investigate the coalescence-induced jumping of mesoscopic water droplets (116.04 nm < R < 290.10 nm).

Although many common phenomena of coalescence-induced jumping of droplets have been analyzed by previous experimental and numerical studies, the process of the energy conversion and the mechanism of enhancing the jumping velocity need to be further exploited. In this paper, the dynamic behaviors and energy conversion efficiency of each stage have been studied for droplets merging on a flat superhydrophobic surface. Besides, the numerical simulations of mesoscopic water droplets of the same size demonstrated that the jumping velocity follows the inertial-capillary scaling law $v_j/v_{ic} \approx 0.105$, enlarging its application range. Moreover, by comparing the coalescence jumping on the substrate with beam and flat substrate, we found that the process of coalescence jumping was largely changed by the beam, which could be considered as a novel jumping mode, and the final jumping velocity can reach $0.7v_{ic}$. The effects of the width and wettability of the beam on the jumping motion of droplets were also considered in our study. Our work provides new insights in the droplet coalescence dynamics and help design anti-icing and anti-frosting surfaces etc.

2. Numerical method

Classical dissipative particle dynamics (DPD) was put forward by Hoogerbrugge and Koelman in 1992 [44]. The basic idea of DPD

follows Newton's equation of motion, for each particle,

$$\frac{d^2\mathbf{r}_i}{dt^2} = \mathbf{F}_i = \mathbf{F}_{i,ext} + \sum_{i \neq j} (\mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R) \quad (1)$$

The mass of each DPD particle is set as 1. The internal force consists of conservative force $\mathbf{F}_{ij}^C$, dissipative force $\mathbf{F}_{ij}^D$ and random force $\mathbf{F}_{ij}^R$, respectively. Owing to that the conservative force of DPD is completely repulsive, classical DPD method is not suitable for the simulation of the gas-liquid interface. Warren [45] imported the conservative force which depends on particles' relative positions and their local densities. This extension of DPD is called many-body DPD (MDPD), where the conservative force is described as:

$$\mathbf{F}_{ij}^C = A\omega^C(r_{ij})\mathbf{e}_{ij} + B(\bar{\rho}_i + \bar{\rho}_j)\omega_d(r_{ij})\mathbf{e}_{ij} \quad (2)$$

$$\omega^C(r_{ij}) = 1 - r_{ij}/r_c \quad (3)$$

$$\omega_d(r_{ij}) = 1 - r_{ij}/r_d \quad (4)$$

$$\bar{\rho}_i(r_{ij}) = \sum_{j \neq i} \omega_\rho(r_{ij}) \quad (5)$$

where $A < 0$ is the coefficient of attractive force, $B > 0$ is the coefficient of density-dependent repulsive force and r_{ij} is the distance between two particles. In that case, the equation of states in the system can have a van der waals loop that enables MDPD to model liquid-vapor interface. The local density is described as Eq. (5). The weight functions $\omega^C(r_{ij})$ and $\omega_d(r_{ij})$ vanish for the distance of particle pairs exceeding the cut-off radius of $r_c = 1.0$ and $r_d = 0.75$, respectively. Therefore, we can define the interaction of various particles by changing the value of A and B. The weight function ω_ρ of local density can be defined by one of the smoothing kernels. In our simulation, we use a bell shaped smoothing function in SPH method by Lucy [46],

$$\omega_\rho(r_{ij}) = \frac{105}{16\pi r_d^3} (1 + \frac{3r_{ij}}{r_d})(1 - \frac{r_{ij}}{r_d})^3 \quad (6)$$

where the weight function vanishes for $r_{ij} > r_d$, and satisfies $\int_0^\infty \omega_\rho(r_{ij}) dr = 1$. Additionally, the dissipative and random forces are matched each other as a thermostat to control the temperature of the system, which are given by

$$\mathbf{F}_{ij}^D = -\gamma\omega^D(r_{ij})(\mathbf{v}_{ij} \cdot \mathbf{e}_{ij})\mathbf{e}_{ij} \quad (7)$$

$$\mathbf{F}_{ij}^R = \gamma\omega^R(r_{ij})\theta_{ij}\mathbf{e}_{ij} \quad (8)$$

$$\omega^D(r_{ij}) = [\omega^R(r_{ij})]^2 = \begin{cases} (1 - r_{ij}/r_c)^2 & (r_{ij} < r_c) \\ 0 & (r_{ij} \geq r_c) \end{cases} \quad (9)$$

where γ is the dissipative coefficient, γ is the random coefficient, $\mathbf{v}_{ij}$ is the relative velocity, ω^D and ω^R are the weight functions, respectively. For a system in a thermodynamic equilibrium state, the temperature T needs to satisfy the fluctuation-dissipation theorem [47] as $\gamma^2 = 2\gamma k_B T$, where k_B is the Boltzmann constant. A modified velocity-Verlet algorithm was employed in present study. The empirical value λ introduced by Groot and Warren [48] is defined as 0.65, and the time step is set as 0.01 in present study.

3. Modeling and parameterizations of droplets

In MDPD simulation, the physical properties are regulated by parameters mentioned in previous section. To study the physical property of MDPD liquid droplet, we defined the liquid droplet with the MDPD parameter listed in Table 1. where the parameter A_{LL} denotes the attractive coefficient and B_{LL} the repulsive coefficient in Eq. (2). The system temperature $k_B T$ is set as 1. Dissipative particle dynamics works in reduced units. The number density ρ can be computed by the relaxation of a spherical liquid droplet. We measure $\rho = 7.49$.

Table 1
Droplet MDPD parameter in reduced units.

Parameter	Symbol	Value
Attractive coefficient	A_{LL}	-50.0
Repulsive coefficient	B_{LL}	25.0
Dissipation parameter	γ	18.0
System temperature	$k_B T$	1.0
Attractive interaction range	r_e	1.0
Repulsive interaction range	r_d	0.75

Table 2
Correspondence of MDPD units and real physical units for water at 20 °C.

Quantities	DPD units	Physical units
$[m]$	1.0	2.11×10^{-22} kg
$[L]$	1.0	1.16×10^{-8} m
$[t]$	1.0	2.08×10^{-10} s
ρ	7.49	998 kg/m ³
γ	14.85	0.072 N/m
μ	11.527	10^{-3} N s/m ²

The simulation of an infinite liquid film was employed to calculate the surface tension. We find $\gamma = 14.85$. With the same density, viscosity was tested by a doubly periodic Poiseuille flow [49], $\mu = 11.53$. To correspond the real physical units, Arienti et al. [50] noticed that for the same properties of surface tension, density and viscosity, the DPD units can be converted as

$$M_{DPD} = L_{DPD}^3 \frac{\rho_{real}}{\rho_{DPD}} \quad (10)$$

$$T_{DPD} = (M_{DPD} \frac{\gamma_{DPD}}{\gamma_{real}})^{0.5} \quad (11)$$

$$\frac{M_{DPD}}{T_{DPD} L_{DPD}} = \frac{\mu_{real}}{\mu_{DPD}} \quad (12)$$

where the subscript “real” means the real liquid physical properties. ρ , μ and γ represent density, viscosity and surface tension, respectively. The properties of water at 20 °C ($\rho = 998$ kg/m³, $\mu = 1.0 \times 10^{-3}$ N s/m², $\gamma = 0.072$ N/m) are corresponded. Three basic physical quantities, mass $[m]$, length $[L]$ and time $[t]$ are derived in Table 2.

An important indicator of wettability is the contact angle. The equilibrium contact angle of a droplet placing on a solid substrate satisfies Young equation [51],

$$\cos \theta_0 = \frac{\gamma_{SV} - \gamma_{SL}}{\gamma_{LV}} \quad (13)$$

The subscript L, V and S refer to liquid, vapor and solid phases, respectively. γ_{SV} , γ_{SL} and γ_{LV} denote the surface tension of solid–vapor interface, solid–liquid interface and liquid–vapor interface, respectively. The contact angle is an important factor in terms of coalescence-induced droplet jumping. To this end, we studied an MDPD system of a liquid droplet on a flat substrate. The parameters of the droplet and the flat substrate are $A_{SS} = A_{LL} = -50.0$, $B = 25.0$, $A_{SL} = -10$. A_{LL} and A_{SS} represent the attractive coefficient for liquid–liquid and solid–solid interactions, respectively. We decrease the attractive interaction between solid and liquid particles so as to reduce the wettability between the droplet and the substrate, which has been adopted in our previous work [52–54]. The equilibrium contact angle was measured as 160°.

4. Nondimensionalization

Some dimensionless numbers are used to characterize coalescence-induced droplet jumping. Ohnesorge number (Oh) which describes the relation of viscous force with inertial and surface tension is given by,

$$Oh = \frac{\mu}{\sqrt{\rho \gamma r}} \quad (14)$$

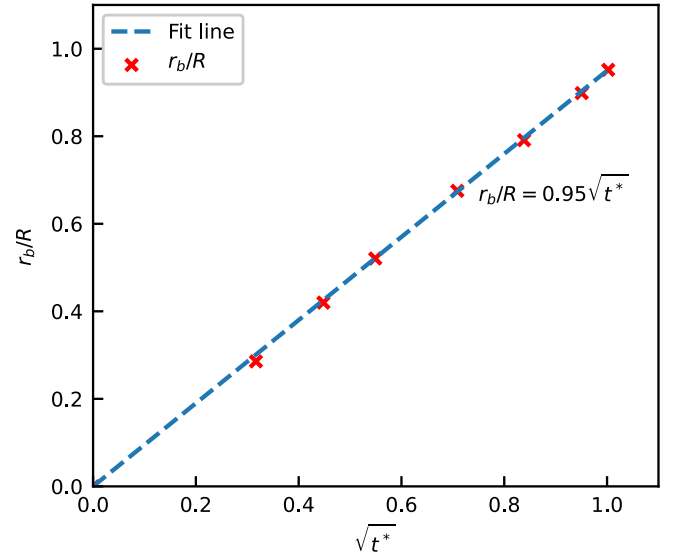


Fig. 1. The radius of liquid bridge as a function of square root of nondimensional time.

where ρ , γ , μ and r are the density, surface tension, viscosity and initial radius of droplet, respectively. Larger Oh number signifies viscosity which plays a more important role in coalescence. If the physical properties of liquid remain unchanged, then Oh number only varies with the radius of the droplet. Thus, the coalescence of smaller droplet leads to the relatively serious internal viscous dissipation, which impedes the jumping motion. In this paper, the scale of water droplet is on the order of a few hundred nanometers, which is much smaller than the capillary length of water $\lambda = \sqrt{\gamma/(\rho g)} = 2.7$ mm. The gravitational effect can be neglected. Hence, the Bond number $Bo = \rho g r^2 / \gamma \approx 0$. Moreover, we defined the nondimensional time t^* and velocity v^* with inertial-capillary time scale t_{ic} and inertial-capillary velocity v_{ic} as follows,

$$t^* = \frac{t}{t_{ic}} = \frac{t}{\sqrt{\frac{\rho r^3}{\gamma}}} \quad v^* = \frac{v}{v_{ic}} = \frac{v}{\sqrt{\frac{\gamma}{\rho r}}} \quad (15)$$

5. Model validation

The liquid bridge forms upon the contact of two droplets. The surface tension drives the expansion motion of liquid bridge rapidly. Eggers et al. [55,56] theoretically reported that the width of liquid bridge follows the scaling law $r_b \propto \sqrt{t}$ when the droplet coalescence is in the inertial regime. Wu et al. [57] experimentally confirmed this scaling law. We nondimensionalize the scaling law as

$$r_b/R = C \sqrt{t^*} \quad (16)$$

where r_b is the radius of liquid bridge, R the initial radius of droplet, C a constant with the order of unity, t^* nondimensional time, respectively. The viscous length scale for water can be calculated as $l_v = \nu^2 \rho / \gamma \approx 14$ nm, where ν is the kinematic viscosity. In our work, $r_b \gg l_v$ represents that the coalescence is carried out in the inertial regime, except for the very beginning of coalescence. We plot the evolution of nondimensional radius of liquid bridge as function of square root of nondimensional time in Fig. 1. The slope of linear fitting line is 0.95, which confirms the scaling law and simultaneously validates our model.

6. Results and discussions

6.1. Coalescence on flat substrate

The details of the simulation are set as Fig. 2. Two droplets are placed on the superhydrophobic solid substrate with a distance of 3

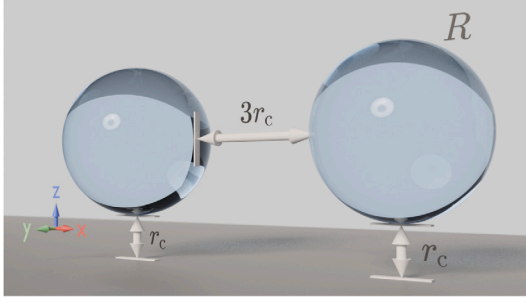


Fig. 2. Illustration of initial simulation of droplet coalescence on the flat substrate.

times cut-off radius. The height of the droplets from the substrate is r_c , where $r_c = 11.6$ nm. After two droplets fully relax to a thermal equilibrium state, a pair of horizontally opposed forces were applied on both droplets for a short time to drive them approaching each other. Since the acceleration generated by the driving force is very small, we consider that the initial droplet coalescence is carried out in a quasi-static state. Previous studies on droplet coalescence process demonstrated that different stages are distinguished by the development of liquid bridge, vertical velocity and vertical acceleration of the mass center of the droplets, where the coalescence jumping process was divided into four or five stages. We consider the two droplets as a group, whose velocity of mass center was calculated every 500 timesteps. Five independent simulations are carried out aiming to reduce the random fluctuation of the results. When two droplets contact and form a tiny liquid bridge, it was thought as the beginning of droplet coalescence. In terms of velocity, the horizontal velocity of one droplet has a sudden increase. Evaluations of nondimensional velocity of the droplet group in z direction and nondimensional velocity in x direction of the droplet locating in the negative x -axis are plotted in Fig. 3, in which the radius of the initial droplet is 197 nm and Oh number is 0.265. Similar phenomena as other work has been observed. The coalescence-jumping process can be divided into four stages and the dynamic morphology of droplets is shown in Fig. 4:

Stage I ($0 < t^* < 0.94$): the liquid bridge expands in y and z direction, until it touch the substrate. The capillary pressure drives two droplets merging and moving to the center. The velocity in x direction of one droplet increases to maximum.

Stage II ($0.94 < t^* < 2.49$), the lower surface of liquid bridge impacts the substrate. The expansion of liquid bridge in z direction is restricted, which provides the upward momentum. During this stage, the velocity in z direction of the droplet group change from negative minimum to positive maximum. In addition, the coalescence droplet oscillates in y and z direction. The velocity in x direction of one droplet reduces. We can consider two droplets as an entirety after stage II.

Stage III ($2.49 < t^* < 4.88$) The area of liquid–solid interface begin to decrease, which indicates that detachment of the coalescence droplet starts. Owing to the adhesive force of solid substrate, the velocity in z direction of coalescence droplet decreases.

Stage IV ($t^* > 4.88$), the coalescence droplet jumps away from the solid substrate. Thus, the coalescence droplet move with the constant final jumping velocity, without any external force acting on it.

The released surface energy can be calculated with the surface area variations of the vapor–liquid interface and the solid–liquid interface.

$$\Delta E_s = \gamma_{LV} \Delta S_{LV} + \Delta S_{LS} (\gamma_{LS} - \gamma_{SV}) \quad (17)$$

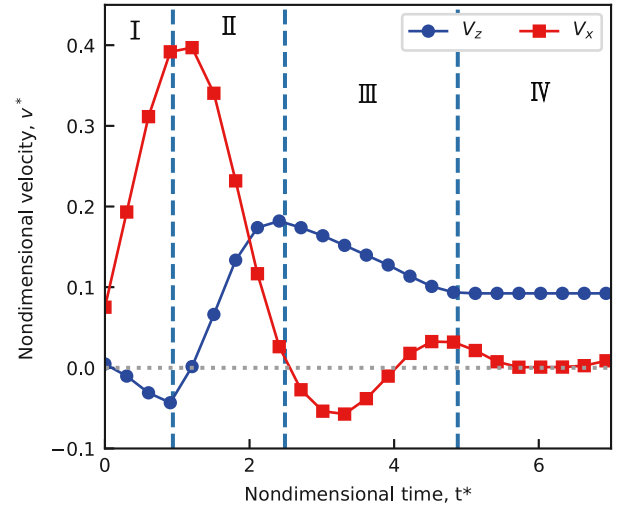


Fig. 3. Evolution of nondimensional velocity, where V_z represents the center of mass velocity of coalescence droplet in z direction, and denotes the center of mass velocity in x direction of the droplet locating at the negative x -axis. ($Oh = 0.265$).

By using Eq. (13), the total released surface energy of two identical droplets coalescence on a flat superhydrophobic surface is given by

$$\Delta E_s = 4\pi R^2 \gamma_{LV} [1 - \cos \theta - (\frac{\cos^3 \theta - 3 \cos \theta + 2}{2})^{\frac{3}{2}} - \frac{1}{2} \cos \theta \sin^2 \theta] \quad (18)$$

where R is the initial radius of the droplet before merging, γ_{LV} is the vapor–liquid interface tension and θ the static contact angle. The energy conservation during droplet coalescence can be expressed as $\Delta E_s = E_{vis} + E_{ad} + E_k$, where E_{vis} represents the viscous dissipation energy, E_{ad} is the work of adhesion, E_k is the kinetic energy of droplets, respectively. The calculation method of the work of adhesion was proposed previously by Athanase Dupré [58]. By combining with Young's equation Eq. (13), the work of adhesion can be shown as

$$E_{ad} = \gamma_{LV} A_{SL} (1 + \cos(\theta)) \quad (19)$$

where A_{SL} denotes the solid–liquid contact area. The nondimensional work of adhesion at t^* moment is defined as

$$E_{ad}(t^*) = \frac{\gamma_{LV} A_{SL}(t^*)(1 + \cos(\theta))}{\Delta E_s} \quad (20)$$

The kinetic energy of droplets can be computed by

$$E_k(t^*) = \begin{cases} \frac{m_1(V_{x1}^2 + V_{z1}^2)}{2} + \frac{m_2(V_{x2}^2 + V_{z2}^2)}{2}, & I \text{ \& II} \\ \frac{(m_1 + m_2)(\bar{V}_x^2 + \bar{V}_z^2)}{2}, & III \text{ \& IV} \end{cases} \quad (21)$$

where m_1, m_2 are the mass of droplets, $V_{x1}, V_{z1}, V_{x2}, V_{z2}$ denote the velocity components of each droplet in x and z direction, $\bar{V}_x, \bar{V}_z$ represent the velocity components of the droplet group in x and z direction. Due to the symmetry of droplet coalescence, the velocity component in the y direction is ignored. During stage I and II, the coalescence has not completed. Thus, the kinetic energy can be computed as the sum of the two droplet's kinetic energy. After that, the coalescence droplet is regarded as an entirety. Most previous researches focus on the jumping velocity and final energy conversion efficiency. The horizontal velocity of each droplet is considered in our work. The energy conversion efficiency η is defined as the ratio of the kinetic energy to released surface energy.

$$\eta = \frac{E_k}{\Delta E_s} \quad (22)$$

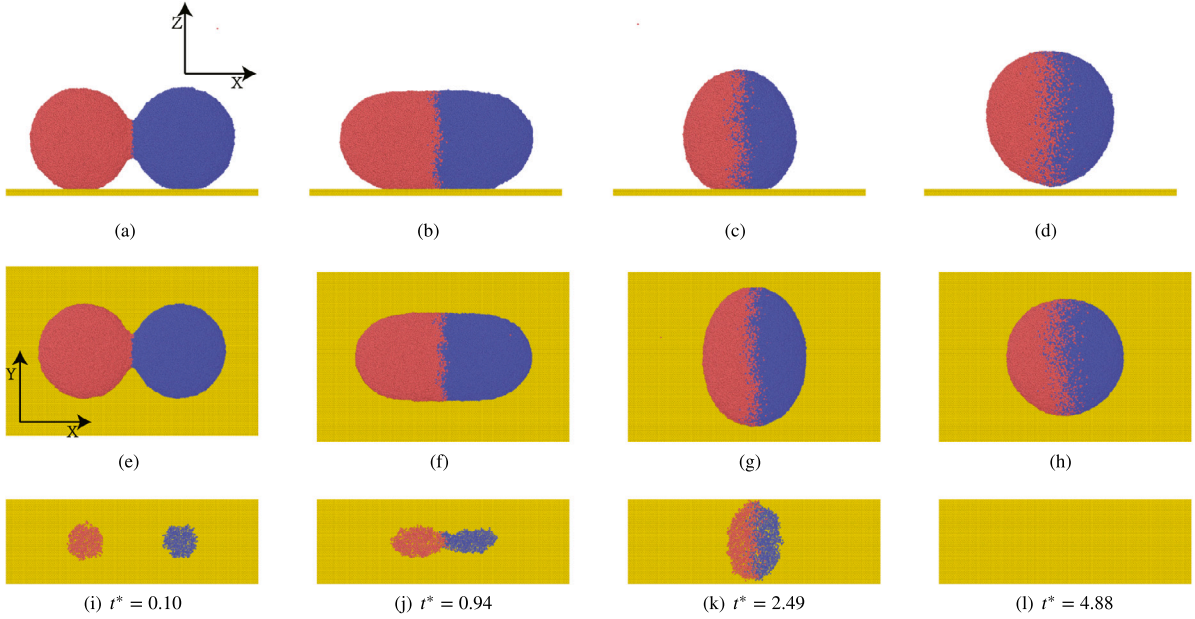


Fig. 4. (a–d) Snapshots in side view, (e–h) snapshots in top view, (i–l) snapshots of liquid–solid interface, showing the dynamic morphology of coalescence droplet at different stage and at the moment of changing stages.

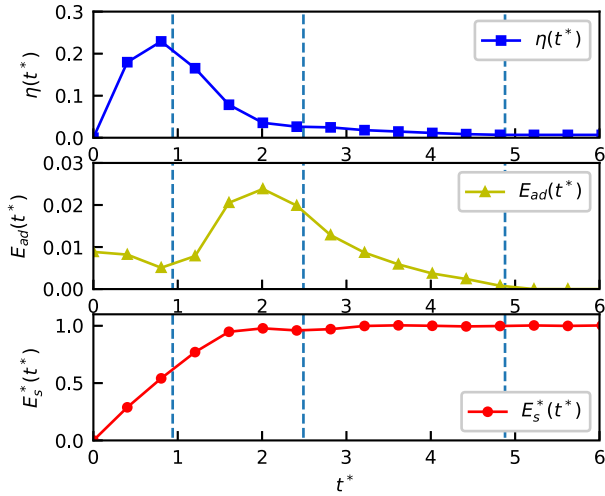


Fig. 5. Evolution of nondimensional work of adhesion, energy conversion efficiency and released rate of surface energy ($Oh = 0.265$).

Therefore, the energy conversion efficiency at t^* moment can be calculated by

$$\eta(t^*) = \frac{E_k(t^*)}{\gamma_{LV}[(S_{LV}(0) - S_{LV}(t^*)) - \cos \theta (S_{LS}(0) - S_{LS}(t^*))]} \quad (23)$$

where $S_{LV}(0) = 4\pi R^2(1 - \cos \theta)$ is the initial area of vapor–liquid interfaces, $S_{LS}(0) = 2\pi R^2 \sin^2 \theta$ is the initial area of liquid–solid interface. The released rate of surface energy can be calculated by

$$E_s^*(t^*) = \frac{\gamma_{LV}[(S_{LV}(0) - S_{LV}(t^*)) - \cos \theta (S_{LS}(0) - S_{LS}(t^*))]}{\Delta E_s} \quad (24)$$

The jump-off time is supposed as t_j^* . Obviously, at the beginning of coalescence, $E^*(0) \approx 0$ and $\eta(0) \approx 0$. Furthermore, the surface energy releases completely at stage IV resulting in the released surface energy rate $E^*(t^* > t_j^*) \approx 1$. The Fig. 5 demonstrates the time evolution of the energy conversion efficiency, nondimensional work of adhesion and released rate of surface energy ($Oh = 0.265$).

The surface energy is substantially released during the stages I and II. After stage II, there is nearly no release of surface energy. The energy conversion efficiency in stage I mainly maintains at about 20%, while 50% surface energy is released. During stage II, despite that the remaining half of the surface energy is released, the kinetic energy decreases due to that most part of the kinetic energy in the x direction is converted into the oscillating energy of droplet in the y direction and is finally dissipated. Owing to a larger liquid–solid contact area in the stage III, the coalesced droplet requires more energy to overcome the work of adhesion. The energy conversion efficiency continues to reduce. The comparison of the kinetic energy and work of adhesion at the beginning of stage III determines whether the droplet can jump away from the substrate. In stage IV, the final energy conversion efficiency is $\eta = 0.007$, which reveals that less than 1% of released surface energy is converted into kinetic energy.

In the present study, the coalescence jumping processes, especially the rule of jumping velocity, of different radius of water droplets ranging from 116.04 nm to 290.10 nm was investigated ($0.218 < Oh < 0.346$). Fig. 6 shows that the coalescence-induced jumping of water droplets can be observed at mesoscopic scale, and the inertial capillary law $v_j/v_{ic} \approx 0.105$ is still applicable. The factor is smaller than 0.126 as pointed out in, where the work of adhesion was ignored when merging on a completely non-wetting substrate. The results demonstrates that the range of Oh number in which the jumping velocity follows the inertial-capillary scaling law is enlarged.

Liu et al. [26] has reported that the departure time of droplet extends as Oh number increases, notably with $0.01 \lesssim Oh \lesssim 0.3$. Each stage durations of different Oh numbers ($0.218 < Oh < 0.346$) were recorded and plotted in Fig. 7. As the influence of the droplet viscosity effect increases with the increasing of Oh number, the coalescence of droplet becomes slow. The duration of liquid bridge expansion in stage I and the duration of overcoming the work of adhesion in stage III augment with the increasing of Oh number. However, the durations of stage II for different Oh numbers are nearly similar with each other, due to that the effect of viscous dissipation is relatively weak.

6.2. Coalescence on substrate with beam

We now consider the droplet coalescence on a substrate with beam. Shown as in Fig. 8, a trapezoidal beam of height h and width d is

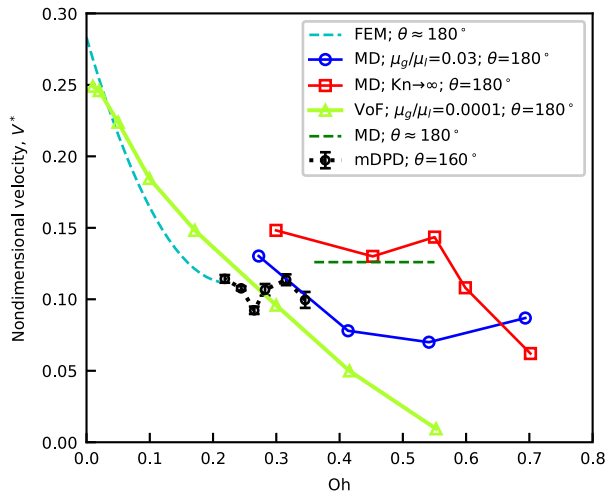


Fig. 6. Comparison of the nondimensional velocity with different methods. Dark green dashed line denotes the simulation results of Ar droplets merging on Au surface with 180° contact angle by using MD method [31]. The wathet dashed line is the results of finite-element-based method [18]. The red, blue and green line are the results of Perumanath et al. [59], in which μ_g/μ_l represents the viscosity ratio of vapor and liquid phases and $Kn \rightarrow \infty$ is viewed as $\mu_g/\mu_l \rightarrow 0$ in VoF simulation.

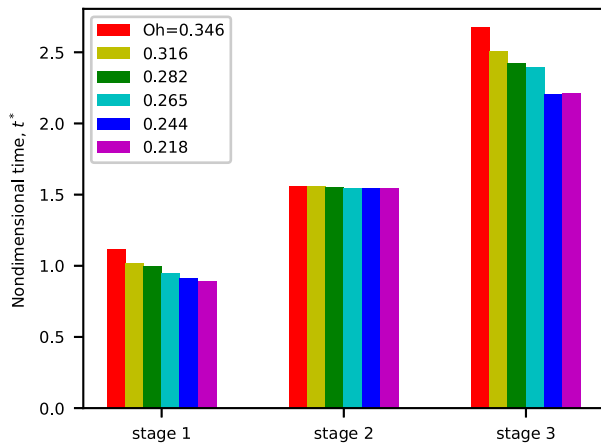


Fig. 7. The duration of different stages with different Oh number.

placed in the middle of two droplets on the substrate. The wettability of beam is the same as the substrate. The lower width of the beam is two units wider than the upper. Thus, during droplet coalescence, droplets would not touch the side of beam. The simulation process is similar to the study of the coalescence-induced jumping on a flat substrate. To compare with flat substrate, we use the same radius droplets of flat substrate system ($Oh = 0.265$).

The effect of the width of beam was investigated firstly. In order to ensure that the beam would not disturb the process before the coalescence of droplets starts, the maximum width of the beam can be estimated by $d \approx 2(R - \sqrt{2h(R-h)})$, where R represents the radius of the droplet. The nondimensional velocities of mass center in z direction for cases with different width of beam were plotted in Fig. 9. For cases of droplets coalescing on a substrate with the same height of superhydrophobic beam, the width of beam has little effect on nondimensional velocity in z direction, and the beam with a larger width results in a small reduction of nondimensional jumping velocity.

Therefore, in order to study the influence of the beam on the process of coalescence-induced jumping of droplets, the upper width of the beam was set as the thickness of a layer of particles ($d = 11.6$ nm).

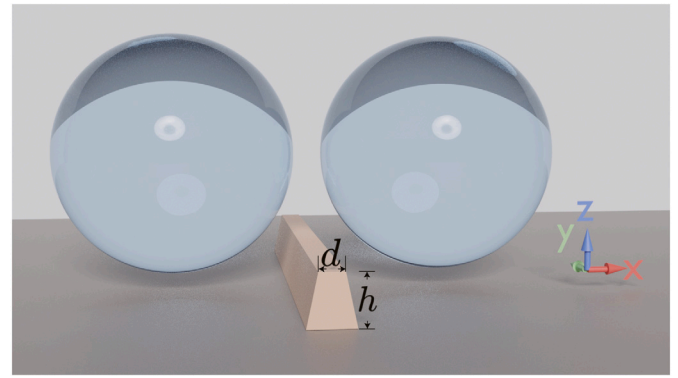


Fig. 8. Illustration of the initial simulation of droplet coalescence on the substrate with a beam of height h .

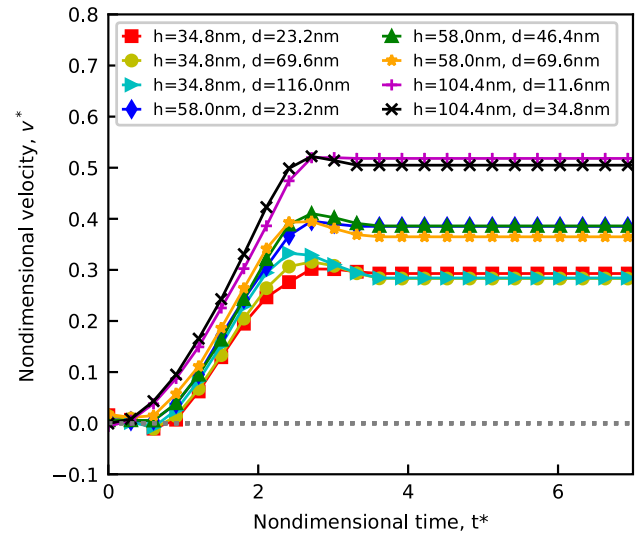


Fig. 9. Comparison of the time evolution of nondimensional velocity in z direction for droplet coalescing on a substrate with a beam of different heights and widths. ($Oh = 0.265$).

The process of jumping on a substrate with a beam resembles to flat substrate. Fig. 10 denotes the comparison of the nondimensional velocity of center of mass between droplet coalescence on substrate with and without beam. Similar to the coalescence-induced jumping on the flat substrate, the liquid bridge develops along y and z directions in stage I ($0 < t^* < 0.50$). However, the ending time of this stage is earlier owing to the beam. At the end of this stage, the velocity in z direction reaches the minimum. During Stage II ($0.50 < t^* < 2.51$), the lower surface of liquid bridge impacts the beam, which hinders the expansion of liquid bridge. The coalescence droplet begins to move upward. The velocity in z direction increases when the droplets continue merging in x direction. As a result, the stage of overcoming the adhesion work of substrate was shifted earlier, when compared with that of merging on flat substrate. At the end of this stage, the droplet departs from the substrate. In stage III ($2.51 < t^* < 3.21$), the droplet suspends in the air, only contacting with the beam. During this stage, the velocity in z direction has a small increase. In stage IV ($t^* > 3.21$), the coalescence droplet moves upward, in the mean time, away from the beam.

The curves in Fig. 10 indicates that the beam has a little impeding effect on the droplet dynamic behavior in x direction. Whereas, the coalescence-induced jumping process is vertically modified by the beam, which prevent the lower surface of droplet from contacting with substrate in stage II. The oscillation of droplet in y direction shown in 11(c) and 11(h) is identical to that of merging on a flat substrate.

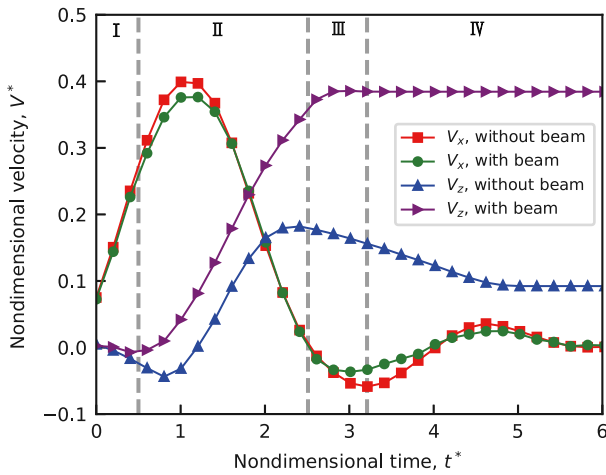


Fig. 10. Comparison of the time evolution of nondimensional velocity between droplet coalescence on a flat solid substrate and on a solid substrate with beam. ($h = 58.02$ nm, $Oh = 0.265$).

Moreover, the form of lower surface of droplet is irregular, indicating that the surface energy is not completely released. Thus, during stage III, the droplet further transforms to a lower surface energy state, and resulting in the acceleration of droplet in z direction.

Obviously, the total released surface energy of coalescence-jumping droplet on a substrate with a beam and without beam is the same. As shown in Fig. 12, the final energy conversion efficiency of coalescence-induced jumping on the substrate with a 58.02 nm high beam is 13.0%, which is enhanced by about 17 times of that on flat substrate. According to the green line, the work of adhesion decreases continuously since the beginning of coalescence on a substrate with a beam. Notably, compared with the curve with no beam, it avoids for the droplet in a state of low kinetic energy at stage III to overcome the strong work of adhesion, thereby improving the conversion efficiency of kinetic energy. Besides, the evolutions of released surface energy of the two cases have a little difference at the end of stage II. The reduction of released surface energy are not observed, meaning that the kinetic energy of the droplet is not converted back to surface energy. Thus, it increases the energy conversion efficiency as well. During stage III, due to that the oscillation in x direction is weak as well as the adhesive effect of substrate disappears, the remaining surface energy is more efficiently converted into the jumping kinetic energy. The curve of energy conversion efficiency shows that the main difference from the coalescence-induced jumping on flat substrate lies in stage II and III. At the beginning of stage II, two droplets are merged, so the kinetic energy conversion efficiency increases at this time. Afterwards, although the kinetic energy of the droplet in the x direction decreases, the kinetic energy of the droplet in the z direction gradually dominates due to the existence of the beam. The energy conversion efficiency increases finally at the end of stage II.

6.3. Effect of the height of beam

We define the nondimensional beam height as $h^* = h_b/R$, where h is the height of beam, R is the initial radius of the droplet. Thus, $h_b \leq R$, otherwise, the droplet will not merge spontaneously. The duration of different stages with different height of beam is plotted in Fig. 13(a). The total jumping time is much smaller than that on flat substrate. As mentioned in the model validation section, the expansion of liquid bridge follows the scaling law of $r_b/R \propto \sqrt{t^*}$. Higher beam evidently leads to an early beginning of stage II. Meanwhile, according Eq. (16), the velocity of liquid bridge impacting the beam can be calculated by

$$v_b = \frac{C^2}{2(1-h^*)} v_{ic} \quad (25)$$

The impacting velocity as well as the counterforce of beam increase with the increasing of the beam height, which results in a larger jumping velocity. The duration of stage II decreases with h^* for smaller h^* , and increases with h^* for larger h^* . Besides, the duration of the stage III increases with the height of the beam. Combining with Fig. 13(b), the droplet is upward accelerated during stage II and III except for the case of tiny beam, since the droplet forms a larger solid-liquid contact area with substrate, resulting in the reduction of jumping velocity in the latter period of stage III. Therefore, the mode of coalescence-induced jumping on substrate with short beam is identical to the flat substrate.

We plot the energy conversion efficiency of coalescence-induced jumping on substrate with different heights of beam in Fig. 14. The maximum energy conversion efficiency, i.e., 44%, could be achieved, when the height of beam equals the radius of the droplet. Evidently, higher beam results in a larger energy conversion efficiency. These curves illustrate that the height of beam has a great influence on energy conversion efficiency. Interestingly, with the beam exceeding a certain height, the energy conversion efficiency re-increases in the late of stage II. When the beam height is reduced to a tiny value, the efficiency curve is substantially similar to the coalescence-induced jumping on a flat substrate. Furthermore, the energy conversion efficiency as a function of the nondimensional height is plotted in Fig. 15. The energy conversion efficiency is linear to the nondimensional height, in which the slope is 0.44, and the intercept is 0.005. It was found in agreement with the results of the experiments by Vahabi et al. [37].

6.4. Effect of the wettability of beam

Influences of the wettability of the beam on the coalescence-induced jumping of droplets were also studied. The wettability of the beam can be changed by adjusting the attractive coefficient of liquid-solid interaction. Fig. 16 shows that the nondimensional velocities of the center of mass in z direction for cases of coalescence-induced jumping of droplets on a beam with height of 58.02 nm and width of 69.6 nm, where the wettability ranges from superhydrophobic to hydrophilic. Different wettability of beam leads to three final behaviors of coalesced droplets, they are the droplet detaching from the beam, standing on the beam and wrapping the beam, respectively.

For the case where the contact angle of beam is 141.8° , the coalesced droplet is still able to detach from the beam. The work of adhesion of beam increases, which results in that the final nondimensional jumping velocity reduces when compared with the case where the contact angle of beam is 160° . Besides, the nondimensional time of detaching from the beam increases. When the contact angle of the beam decreases to 119.8° , the maximum nondimensional velocity of the droplet in the z direction continues to decrease, which is not enough to detach from the upper surface of beam. The coalesced droplet is pulled back by the force of adhesion of the beam, and oscillates in z direction on the beam. Finally, the coalesced droplet stands on the beam, shown as Fig. 17. For coalescence-induced jumping of droplets on substrate with a hydrophobic beam, the jumping process follows the four stages that have mentioned in the previous section. However, the only difference is whether the droplet finally jumps away from the beam. For the cases of hydrophilic beams, the nondimensional velocity in z direction increases firstly like the cases with hydrophobic beam. Due to the hydrophilic beam, droplets are absorbed by the side surfaces of beam during the process of coalescence. The nondimensional velocity in z direction decreases and then oscillates. In the end, the coalesced droplet wraps the beam, and stands on the substrate, shown as Fig. 18.

The final behaviors of droplets jumping on the substrate with different geometry and wettability of beam are listed in Table 3. The width could change the final behaviors of the coalesced droplet when the height and the wettability of the hydrophobic beam are fixed (CA = 141.8° and $h = 34.8$ nm).

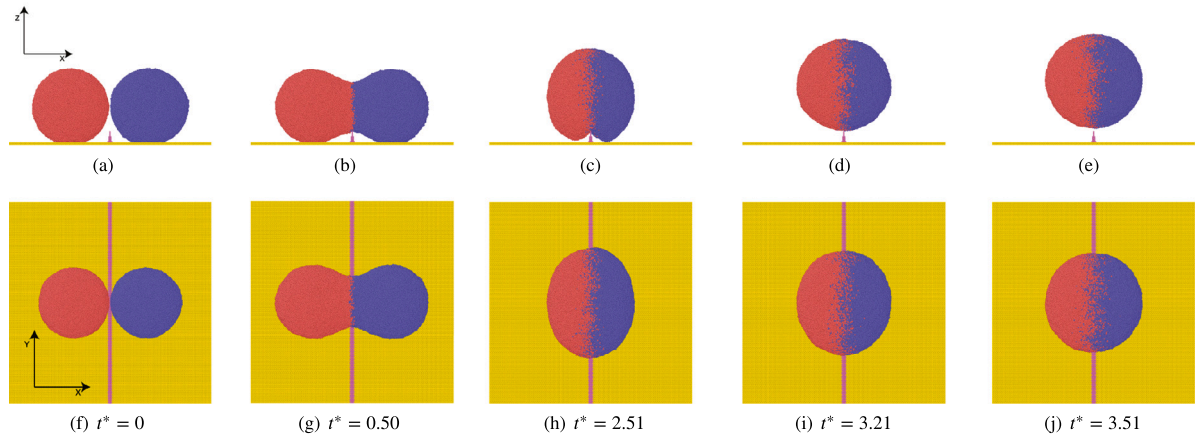


Fig. 11. (a–e) Snapshots in side view, (f–j) snapshots in top view, showing the dynamic morphology of coalescence droplet at different stages and at the moment of changing stages.

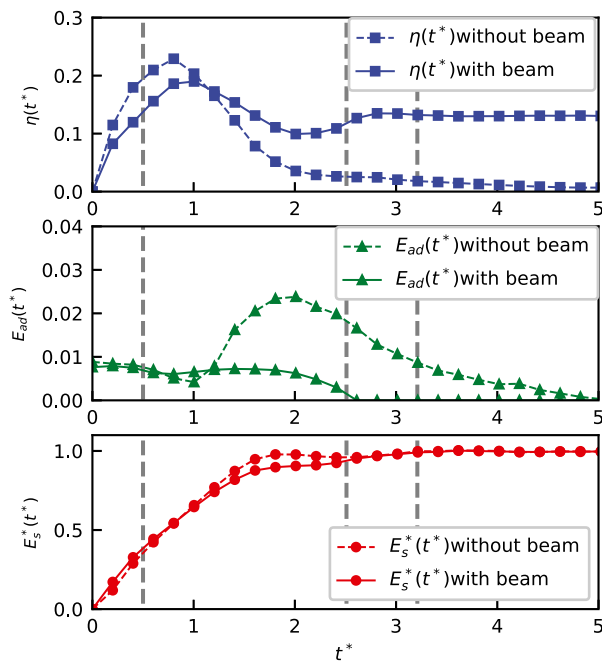


Fig. 12. Comparison of the time evolution of nondimensional work of adhesion, energy conversion efficiency and surface energy released rate between droplet coalescence on a flat solid substrate and on a solid substrate with beam ($h = 58.02$ nm, $Oh = 0.265$).

Table 3

The final behaviors of droplets jumping on substrates with different geometries and wettability of beam.

CA	Height, h	Width, d	Final behavior
141.8°	34.8 nm	92.8 nm	Jumping
141.8°	34.8 nm	139.2 nm	Standing
141.8°	58.0 nm	46.4 nm	Jumping
141.8°	58.0 nm	69.6 nm	Jumping
141.8°	58.0 nm	92.8 nm	Jumping
85.8°	34.8 nm	92.8 nm	Wrapping
85.8°	34.8 nm	139.2 nm	Wrapping
85.8°	58.0 nm	46.4 nm	Wrapping
85.8°	58.0 nm	69.6 nm	Wrapping
85.8°	58.0 nm	92.8 nm	Wrapping

7. Conclusions

In summary, the behaviors of coalescence jumping of two mesoscale water droplets on a flat substrate and on a substrate with beam were

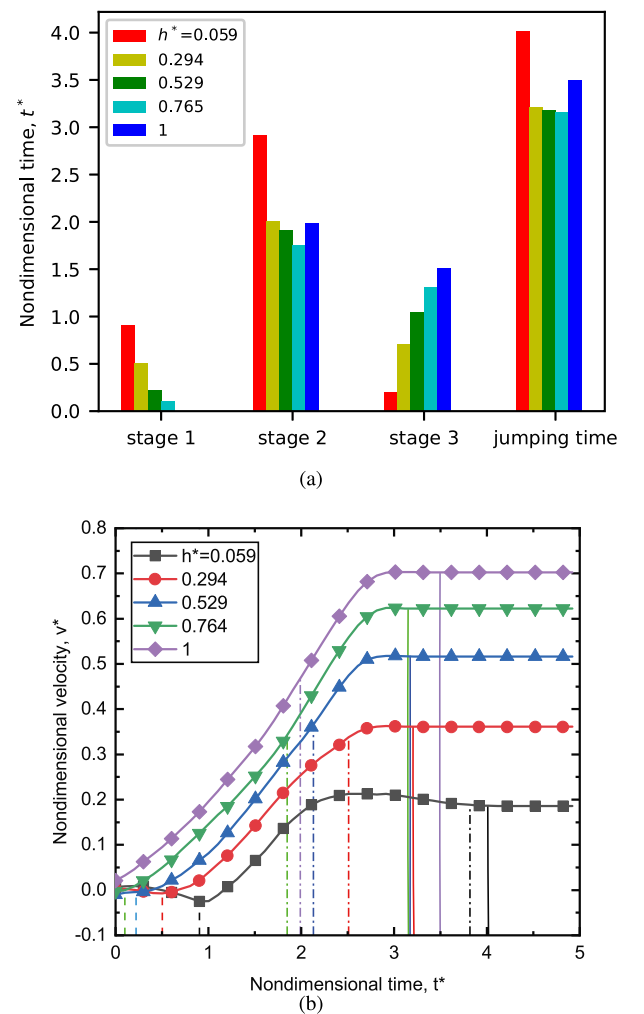


Fig. 13. (a) Duration of different stages with different h^* ($Oh = 0.265$), (b) Time evolution of nondimensional velocity in z direction of droplet coalescence on substrate with different height of beam. Dashed line, dashed dot line and solid line denote the stage (I–II, II–III and III–IV) transition time point.

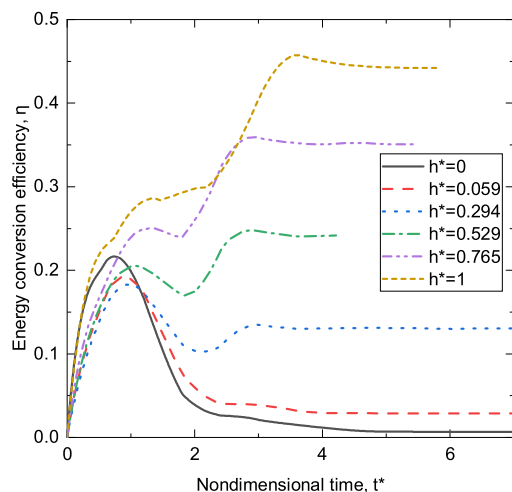


Fig. 14. Time evolution of kinetic energy conversion efficiency of different height of beam.

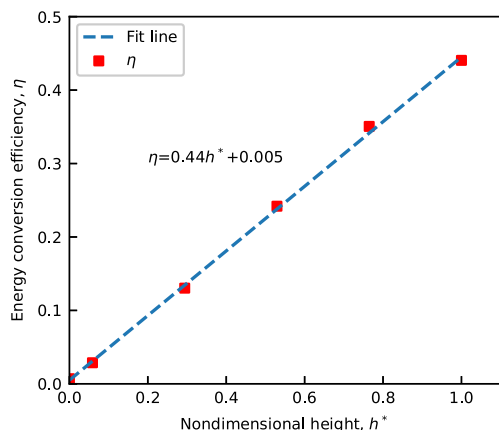


Fig. 15. Energy conversion efficiency as a function of nondimensional height.

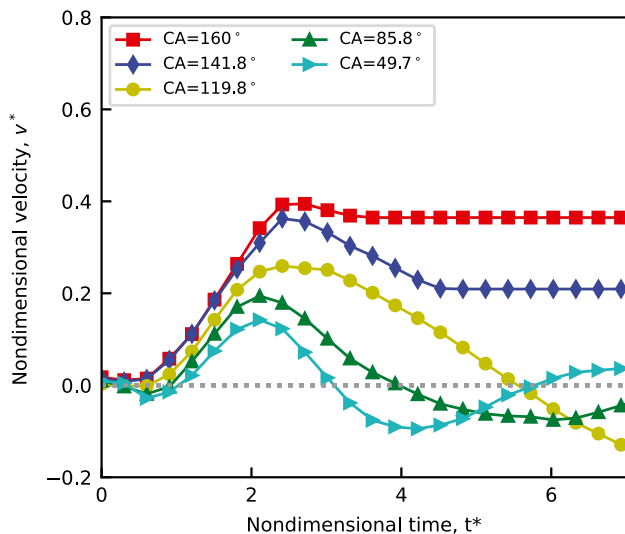


Fig. 16. Comparison of the time evolution of nondimensional velocity of droplet coalescence on substrate with different wettability of beam. ($h = 58.02$ nm, $d = 69.6$ nm, $Oh = 0.265$).

acceleration (II), detaching the adhesion of substrate (III), jumping away from the substrate (IV). The surface energy releases completely

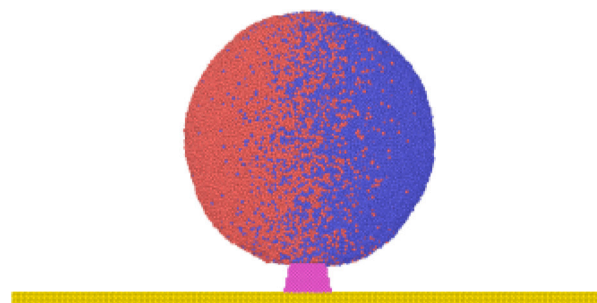


Fig. 17. The final snapshot of droplets coalescing on a substrate with a hydrophobic beam ($h = 58.02$ nm, $d = 69.6$ nm, $Oh = 0.265$, $CA = 119.8^\circ$).

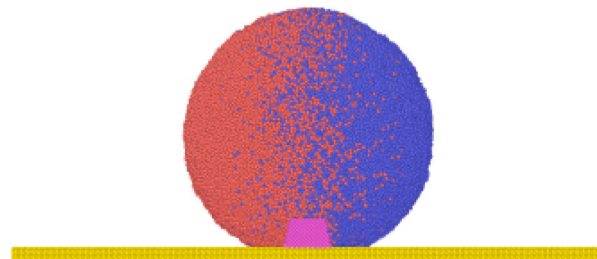


Fig. 18. The final snapshot of droplets coalescing on a substrate with a hydrophilic beam ($h = 58.02$ nm, $d = 69.6$ nm, $Oh = 0.265$, $CA = 85.8^\circ$).

after stage II. The kinetic energy conversion efficiency increase in stage I to reach maximum and decrease in stage II and III. Finally, only around 1% released surface energy converts to jumping kinetic energy. The ratio of jumping velocity to inertial-capillary velocity is approximately fixed at 0.105 for droplet radius ranging from 116.04 nm to 290.10 nm, meaning that the inertial-capillary scaling is still applicable.

The jumping velocity can be enhanced by placing a beam in the middle of the two droplets. In vertical direction, the stage of droplet detaching the substrate and accelerating upward are integrated into stage II for the droplets merging on a substrate with beam. The beam changes the process of coalescence jumping. For droplets merging on substrate without beam, the merging occurs prior to the jumping. While for droplets merging on substrate with a beam, the jumping is accompanied by the merging process, resulting in a longer phase of upward acceleration of the coalesced droplet, a lower adhesion work and a secondary acceleration by releasing the surface energy stored in the liquid-vapor interface around the beam. The maximum energy conversion efficiency during droplet coalescence-induced jumping on substrates with a beam could reach 44%, which is enhanced around 60 times higher than that on flat substrates. The jumping velocity increases with the increasing of the beam height in the range from zero to the radius of droplet. We found that the conversion rate is linear with the height of the beam. Due to that the different wettability of the beam could result in three final behaviors of coalesced droplets, i.e., detaching from the beam, standing on the beam and wrapping the beam, respectively. Moreover, the effect of width of the beam on the jumping of droplets depends on the wettability of beam. For superhydrophobic beams, larger width results in a small reduction of jumping velocity. However, for hydrophobic beams, the final behavior of merged droplet could be changed by the width of beam.

CRediT authorship contribution statement

Zhiyuan Yu: Conceptualization, Methodology, Software, Writing – original draft. **Kaixuan Zhang:** Data curation, Writing – original draft. **Jiayi Zhao:** Funding acquisition. **Shuo Chen:** Writing – review & editing. **Chensen Lin:** Visualization. **Yang Liu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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